

No Trading Strategy Can Win on Every Price Path: Computability, Randomness, and the Limits of Universal Trading

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Can one algorithm make money whatever the market does? The recurring configuration is a trader on one side and a market on the other. A responsive computable market can use the trader’s own output to build a diagonal counterpath. If market rules can emulate universal computation, deciding their future price events would solve the Halting Problem. At the opposite pole, a passive Martin-Löf-random record defeats every effective fair-game betting test. A large or public trader may also change prices through impact and strategic response. Repeatable success therefore requires some exploitable restriction—trend, information, a risk premium, or another structural insight. No-arbitrage, No-Free-Lunch averaging, and time-reversal stress tests sharpen these limits; conditional and relative edges remain possible.

I. INTRODUCTION

Throughout this paper the elementary configuration has two sides: a *trader* and a *market*. At each date the trader converts the observed price history into a position; the market supplies the next price, and their interaction determines the gain. Depending on the model, “market” may mean a realized price path, a probability law over paths, or a responsive rule that observes the trader’s action. This organizing picture is not a claim that every financial model is literally a two-person zero-sum game. It keeps visible which powers and restrictions are assigned to each side.

The universal claim examined here has the quantifier order

$$\exists \sigma \forall \mathcal{E} : \quad \sigma \text{ wins against the market } \mathcal{E}. \quad (1)$$

The computational no-go result reverses it:

$$\forall \sigma \exists \mathcal{E}_\sigma : \quad \sigma \text{ does not win against } \mathcal{E}_\sigma. \quad (2)$$

Here the trader σ is a total deterministic program. After it announces long, short, or flat, a responsive computable market can move down, up, or nowhere, respectively. The countermarket is made from the trader’s own output by reversing that output’s intended effect: this is the diagonal step. A trader that never takes a position—that is, does not trade at all—earns zero rather than a strict loss. The precise conclusion is therefore failure to guarantee a strictly positive gain.

Universal computation creates a second, related obstruction on the market side. If admissible market generators can simulate arbitrary Turing machines, a future price event can encode whether a computation halts. A program that could infer every such rule well enough to decide every encoded event would therefore solve the Halting Problem. This is the concrete rule-inference barrier developed in Sec. II. Gold’s identification-in-the-limit

framework and later work on unsolvable inductive inference place the same boundary in a broader learning-theoretic setting [1–5]. The diagonal countermarket and the Halting reduction share a self-referential ancestry, but they prove different statements and have different quantifiers.

At the opposite pole, the market need not observe the trader at all. A passive binary price record may be Martin-Löf random relative to a specified computable fair measure. Operationally, it then passes every effective null test and permits no lower-semicomputable nonnegative fair-game capital process to become unbounded. This does not mean that no individual move can be guessed, or that finite lucky profits are impossible. It means that an effective trader cannot extract a sustained fair-game advantage from that path. Algorithmic randomness and deterministic chaos are not synonyms.

Between a passive random market and a perfectly responsive countermarket lies the economically important case in which the trader changes the market. A large, crowded, or publicly known strategy consumes liquidity, moves quotes, reveals information, and attracts imitation or opposition. The price record generated after deployment is then partly a function of the strategy itself. This resembles an invasive classical measurement: the intervention changes the process being measured. It is an endogeneity mechanism, not a physical measurement postulate and not a theorem that every large trader must lose.

The practical conclusion is therefore conditional rather than nihilistic. Repeatable success requires an *inductive bias*: a restriction or asymmetry that the trader understands and the unrestricted market class does not assume away. A trend follower posits persistence, a contrarian posits mean reversion, an option seller relies on a risk premium, and an informed trader relies on information not yet fully reflected in prices. Such insights may work in a particular environment. What cannot be supplied is an insight-free algorithm that wins over every admissible environment.

Time reversal is the simplest transparent challenge to

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one such claim: hold the trader fixed, reverse a finite market record, and run the trader causally from the new beginning. It is one member of a countable family of computably specified stress transformations. This countability echoes the effective-test viewpoint of algorithmic randomness, but a reversal stress test is not itself a Martin-Löf test, because it carries no effective probability bound.

Here a strategy is called *universal* only if, after discounting by a specified numeraire, it produces a strictly positive terminal gain from zero initial capital on every admissible market trajectory. The zero-cost, self-financing, and discounted-gain qualifications are essential: a positive initial investment in a bank account has positive terminal wealth, but it is not an arbitrage and does not outperform its financing benchmark.

The mathematical study of unpredictable prices begins with Bachelier's 1900 thesis [6], continues through Samuelson's martingale formulation [7], and informs Fama's Efficient Market Hypothesis (EMH) [8]. The EMH is an economic claim about the incorporation of information into prices; it is not, by itself, a theorem that no individual strategy can win unconditionally.

Section II gives the diagonal counterpath, two explicit Halting reductions, and the Martin-Löf, computable, and Schnorr randomness hierarchy. Section III develops time reversal and other computable stress transformations. Sections IV and V give the separate no-arbitrage and No-Free-Lunch boundaries. The remaining sections delimit the social-choice analogy, analyze the Wheel Options Strategy, distinguish Cover's benchmark-relative universality, and state the honest conditional scope of strategies that work *for all practical purposes* (FAPP).

II. THE TRADER AND THE MARKET: COMPUTABILITY AND RANDOMNESS

We now formalize the trader-market pair. The trader is a program M_σ mapping each observed finite price history to a position; the market is the environment that supplies the next price. Three specifications of the market side lead to three different boundaries: a responsive computable market yields a tailored counterpath, a computationally universal class of market generators makes unrestricted future-event inference undecidable, and a passive algorithmically random record defeats effective fair betting.

A. A Responsive Market: The Diagonal Counterpath

First discard the assumption of an exogenous price process and let the market react after observing the trader's current position. This is a worst-case, perfect-information model. Let M_σ be a *total* computable map from every finite computable price history H_t to a rational position w_t . Totality models the practical require-

ment that a deployed algorithm return an action before a deadline.

Theorem 1 (Computable Adaptive Counterpath). *For every total deterministic computable strategy M_σ and every finite horizon T , there exists a positive computable price path P_{adv} on which its self-financing gain*

$$G_T = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t)$$

is non-positive. It is strictly negative if the strategy takes a nonzero position at least once.

Proof. Choose computable $P_0 > 0$ and rational $0 < \epsilon < 1$. At time t , compute $w_t = M_\sigma(H_t)$ and set

$$P_{t+1} = \begin{cases} P_t(1 - \epsilon), & w_t > 0, \\ P_t(1 + \epsilon), & w_t < 0, \\ P_t, & w_t = 0. \end{cases} \quad (3)$$

The resulting prices remain positive and computable. In each period, $w_t(P_{t+1} - P_t) \leq 0$, with strict inequality when $w_t \neq 0$. Summing proves the claim. $\square$

The construction is diagonal in the following precise operational sense: the market takes the strategy's own current output and applies a fixed-point-free sign reversal to it. Long is answered by down, short by up, and neutrality by zero. The candidate rule is therefore defeated on an input made from its own answer, just as a diagonal argument defeats a purported universal rule by turning that rule against itself. No enumeration of all machines is required. Its quantifiers are $\forall M_\sigma \exists P_{\text{adv}}$: the counterpath may depend on the strategy. There need not be one finite path on which every strategy loses; for example, a permanently long and a permanently short position cannot both lose on the same one-step price move.

B. Universal Computation and the Halting Barrier

Theorem 2 (Undecidability of Eventual Profit). *Fix the positive computable price path $P_t = t + 1$, $t \in \mathbb{N}$. There is no algorithm that, given the code of an arbitrary total computable trading program M_σ , always decides whether*

$$\exists T \geq 1 : \quad G_T(M_\sigma; P) = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t) > 0.$$

Proof. Assume such a profit certifier exists. Given an arbitrary Turing machine A and input x , construct a trading program $M_{A,x}$ as follows. At date t , simulate $A(x)$ for $t + 1$ steps. If it has halted within those steps, set $w_t = 1$; otherwise set $w_t = 0$. Every date requires only a finite simulation, so $M_{A,x}$ is total. Along $P_t = t + 1$, every long position earns one unit in the next period. Hence $G_T > 0$ for some T if and only if $A(x)$ eventually halts. A certifier for eventual trading profit would therefore decide the Halting Problem, which is impossible [9]. $\square$

The reduction can also be placed directly on the market side, which is the form most closely connected to general rule inference.

Theorem 3 (Undecidability of a Market Event). *There is no algorithm that, given the code of an arbitrary total computable positive-price generator G , always decides whether its generated path P^G ever satisfies $P_t^G > P_0^G$.*

Proof. Given a Turing machine A and input x , define the total generator by $P_0^{A,x} = 1$ and, for $t \geq 1$,

$$P_t^{A,x} = 1 + \mathbf{1}\{A(x) \text{ halts within } t \text{ steps}\}, \quad t \geq 1. \quad (4)$$

Each quoted price requires only a finite t -step simulation, so the generator is total and computable. Its initial price is one, and it ever quotes a higher price if and only if $A(x)$ halts. A universal decider for the stated market event would therefore decide the Halting Problem. $\square$

Theorem 3 makes the assumption of universal computation precise: the admissible language of market generators must be rich enough to emulate an arbitrary machine and expose the result through an observable price event. Under that condition, no computable rule-inference procedure can uniformly settle all future events of all such markets. This is a concrete market version of the limits studied in identification in the limit and in unsolvable inductive inference [1–5, 10]. It does not deny that restricted parametric, stochastic, or otherwise structured model classes can be learned.

The three computational claims are related but not identical. Theorem 1 has quantifiers $\forall M_\sigma \exists P_{\text{adv}}$ and constructs a countermarket from the action of a fixed trader; that construction only needs enough computation to run that trader. Theorem 2 embeds an arbitrary computation in the trader while fixing a very simple rising market. Theorem 3 embeds it in the market and rules out a general future-event inference procedure. For a specified finite horizon, by contrast, total programs can simply be run and their realized gains and prices computed.

Corollary 4 (Maximin Value). *In the finite-horizon game of Theorem 1, if the neutral strategy is allowed, then*

$$\sup_{M_\sigma} \inf_{P_{\text{adv}}} G_T(M_\sigma, P_{\text{adv}}) = 0.$$

Proof. The neutral strategy guarantees zero, while Theorem 1 gives every strategy a counterpath with gain at most zero. $\square$

Proposition 5 (Private Randomization Does Not Create a Guarantee). *Suppose a randomized trader’s current coin toss is hidden from the environment. There still exists a passive computable distribution of next returns under which every integrable strategy has conditional expected gain zero.*

Proof. Let the next price increment be $+\epsilon P_t$ or $-\epsilon P_t$ with equal probability, independently of the trader’s private

draw. Conditional on the current information and position, $\mathbb{E}[w_t(P_{t+1} - P_t) \mid \mathcal{F}_t] = 0$. Summing gives expected gain zero. $\square$

Randomization therefore prevents a code-observing opponent from mechanically opposing the unseen current draw, but it does not yield strictly positive expected gain in every environment.

C. The Responsive Market as a Game

Theorem 1 can be read as a finite perfect-information zero-sum game: the trader announces w_t , the environment chooses the next return, and the terminal payoff is G_T . Its value is zero by Corollary 4; backward induction, not a diagonal enumeration, identifies the environment’s response. For infinite perfect-information games, Borel determinacy guarantees that one player has a pure winning strategy when the winning set is Borel [11], but the theorem does not identify the winning player without analysis of the payoff set. Nor does it cover a trader’s hidden random bits: that is an imperfect-information game whose relevant conclusion here is the mixed value zero in Proposition 5.

D. A Passive Market: Martin-Löf Randomness

The market need not observe or answer the trader. At the opposite pole, suppose an effective coding of price directions is Martin-Löf random relative to a specified computable fair measure. Here “random” means algorithmically random, not merely sensitive dependence in deterministic chaos. Such a sequence may still be guessed correctly on many individual dates. The exact claim is that it passes every effective null test, or equivalently that no lower-semicomputable nonnegative fair-game capital process becomes unbounded on it.

In the fair binary market, a nonnegative capital process $K: \{0, 1\}^{<\mathbb{N}} \rightarrow \mathbb{R}_{\geq 0}$ is a martingale when

$$K(s) = \frac{K(s0) + K(s1)}{2}. \quad (5)$$

Theorem 6 (Ville’s Inequality). *If $K_0 = 1$ is a nonnegative martingale under the fair-coin measure, then for every $\lambda > 0$,*

$$\mathbb{P}\left(\sup_{n \geq 0} K_n \geq \lambda\right) \leq \frac{1}{\lambda}. \quad (6)$$

Ville’s inequality follows by stopping the nonnegative capital process when it first crosses λ and applying optional stopping, followed by a limit [12]. In particular, the set on which a fixed nonnegative martingale becomes unbounded has probability zero.

To state the effective refinement correctly, let λ denote fair-coin measure on Cantor space $\{0, 1\}^{\mathbb{N}}$. A Martin-Löf

test is a uniformly computably enumerable sequence of open sets $(U_k)_{k \geq 1}$ satisfying $\lambda(U_k) \leq 2^{-k}$. A sequence is *Martin-Löf random* if it avoids $\bigcap_k U_k$ for every such test [13]. A sequence is *computably random* if no total computable nonnegative martingale succeeds on it by becoming unbounded. A *Schnorr-random* sequence passes every Martin-Löf test whose component measures are uniformly computable. These are distinct notions, with strict implications

$$\begin{aligned} \text{Martin-Löf random} &\implies \text{computably random} \\ &\implies \text{Schnorr random.} \end{aligned} \quad (7)$$

Neither converse holds [14–16].

Theorem 7 (Effective Martingale Barrier). *A binary sequence x is Martin-Löf random if and only if no lower-semicomputable nonnegative supermartingale succeeds on x by unbounded capital. Consequently, the success set of each such betting strategy is effectively λ -null, and the set of paths on which all of them remain bounded has fair-coin measure one.*

Proof. Normalize a lower-semicomputable nonnegative supermartingale d by $d(\emptyset) \leq 1$, and let

$$U_k = \{x : \exists n, d(x \upharpoonright n) \geq 2^k\}.$$

Lower semicomputability makes U_k computably enumerable and open, while Ville’s inequality gives $\lambda(U_k) \leq 2^{-k}$. Hence (U_k) is a Martin-Löf test and no Martin-Löf-random sequence permits d to become unbounded. Conversely, every Martin-Löf test can be converted into a lower-semicomputable supermartingale that succeeds on its intersection; this is the martingale characterization of Martin-Löf randomness [15, 17]. $\square$

The same construction extends from λ to any computable probability measure μ with the supermartingale condition and cylinder probabilities defined relative to μ . This is the effective analogue most directly applicable to a computable martingale measure.

Theorem 7 contains the proposed “typical path” claim in its correct form. Under a computable fair measure, a measure-one class of paths simultaneously blocks unbounded success by every effective nonnegative fair-game capital process. It does *not* say that every such path gives every strategy a finite loss, nor does it apply unchanged to strategies with unrestricted borrowing or to non-martingale physical measures. There is also a direct prediction reading. If a total computable next-bit predictor maintained a long-run accuracy strictly above $1/2$, then betting a sufficiently small fixed fraction of capital on each prediction would produce an unbounded computable martingale. Thus no such sustained directional advantage is possible on a Martin-Löf-random fair-coin path, although finite streaks and isolated correct forecasts remain possible.

Theorem 1 is different: its market is tailored to one deterministic trader and gives that trader a finite non-positive gain. A Martin-Löf-random market need not react to the trader and simultaneously passes all effective tests. The former is a worst-case construction; the latter is a typical-path statement relative to a specified computable measure. Game-theoretic probability develops this pathwise betting viewpoint systematically [18].

E. What Rice’s Theorem Does and Does Not Say

Rice’s theorem states that every nontrivial extensional property of arbitrary partial computable functions is undecidable [19]. It therefore limits fully general program certification; even deciding whether arbitrary code is total is impossible in general [9]. It does *not* imply that the loss-producing inputs of a fixed total finite-horizon strategy are undecidable. On a specified finite computable path, such a program can be simulated and its gain calculated. No probability claim about market failure sets follows from Rice’s theorem alone. Theorem 2 supplies the exact statement needed here by an explicit Halting-Problem reduction: it is the unbounded question of *eventual* profit, not the evaluation of a completed finite trade record, that is undecidable in general.

III. THE TIME-REVERSAL STRESS TEST

The trader–market viewpoint suggests holding the trader fixed while transforming the market record. Among computably specified stress transformations, time reversal is perhaps the simplest.

Let $\mathcal{M}$ be a space of strictly positive paths $P: [0, T] \rightarrow \mathbb{R}_{>0}^n$. The time reversal of P is

$$\tilde{P}(t) = P(T - t), \quad t \in [0, T]. \quad (8)$$

For a causal strategy σ , let $\Pi_T[\sigma; P]$ denote its terminal discounted gain from zero initial capital when evaluated on P .

Definition 1 (Pathwise Universal Strategy). A causal strategy σ is *pathwise universal* on $\mathcal{M}$ if it is self-financing and admissible and satisfies $\Pi_T[\sigma; P] > 0$ for every $P \in \mathcal{M}$.

Criterion 1 (Reversal Falsification Test). Suppose $\mathcal{M}$ is closed under time reversal. If there exists $P \in \mathcal{M}$ such that

$$\min\{\Pi_T[\sigma; P], \Pi_T[\sigma; \tilde{P}]\} \leq 0, \quad (9)$$

then σ is not pathwise universal on $\mathcal{M}$.

Proof. Both P and $\tilde{P}$ belong to $\mathcal{M}$. A non-positive gain on either member of the pair contradicts Definition 1. $\square$

Criterion 1 is deliberately a falsification device, not an independent impossibility theorem. Reversing a profitable path does not in general negate the profit of a causal strategy, because the strategy can take different positions on the reversed history. The value of the test is that it generates a paired stress scenario whenever the modeled path class is reversal-closed.

Remark 1 (Computable Stress Tests Beyond Reversal). Time reversal is only the first member of a much larger battery. Depending on the admissible market class, a computable probe can reverse return signs, splice trend and anti-trend regimes, insert jumps or observation delays, rescale volatility, or add financing and transaction costs. Because programs have finite descriptions, computably specified probes form an at most countable family, and countably infinite test batteries can be constructed. Each transformed record must still belong to the market class against which the strategy's claim is made.

The word “test” is used here in the empirical sense of stress and falsification. A Martin-Löf test is a different technical object: a uniformly computably enumerable sequence of open sets with effective measure bounds. The shared idea is to confront an algorithm with many effective challenges; time reversal by itself neither defines a null set nor certifies randomness. Passing it establishes limited robustness, never universality.

Remark 2 (Filtrations and Time Reversal). In stochastic finance, the set of admissible trajectories is not automatically closed under time reversal. The process $P(T-t)$ is generally not adapted to the original forward filtration, and its semimartingale property under a reversed filtration requires additional hypotheses; time reversal of diffusions is a separate theorem, not a formal substitution $t \mapsto T-t$ [20].

A simple Black–Scholes calculation illustrates both the useful case and its limit. If

$$X_t = \log(S_t/S_0) = mt + \sigma W_t,$$

then the reversed increment $Y_t = X_{T-t} - X_T$, relative to its reversed filtration, has the law of a Brownian motion with drift $-m$ and volatility σ . Thus a log-price model class containing both drift signs is closed in law under reversed increments. Under the physical measure, $m = \mu - \sigma^2/2$; under a risk-neutral measure the discounted stock is a martingale, but its logarithm still has Itô drift $-\sigma^2/2$. Consequently, neither an ELMM nor a driftless arithmetic Brownian special case establishes detailed balance for a general financial model. The stress test remains pathwise; the rigorous probabilistic constraint is given next.

IV. THE NO-ARBITRAGE FOUNDATION

Here the market is no longer an opponent choosing a response after seeing the trader's action. Instead, finan-

cial admissibility and no-arbitrage delimit which trader–market pairs are coherent.

A. Discounted Gains and Admissibility

Let the market be defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$. Let $B_t > 0$ be a numeraire and S_t the vector of traded asset prices. Write $\bar{S}_t = S_t/B_t$ for discounted prices. A predictable, $\bar{S}$ -integrable strategy H has discounted self-financing gain

$$G_t = (H \cdot \bar{S})_t, \quad G_0 = 0. \quad (10)$$

It is *admissible* if there is a finite constant a such that

$$G_t \geq -a \quad \text{for all } t \in [0, T], \quad \mathbb{P}\text{-a.s.} \quad (11)$$

This uniform lower bound rules out doubling systems financed by unbounded credit.

Theorem 8 (Universality Implies Classical Arbitrage). *If an admissible self-financing strategy H with zero initial capital satisfies $G_T(\omega) > 0$ for every $\omega \in \Omega$, then H is a classical arbitrage.*

Proof. The pathwise assumption implies $G_T \geq 0$ almost surely and $\mathbb{P}(G_T > 0) = 1$. Together with zero initial cost, self-financing, and admissibility, these are stronger than the usual classical-arbitrage conditions $\mathbb{P}(G_T \geq 0) = 1$ and $\mathbb{P}(G_T > 0) > 0$. $\square$

Remark 3 (Terminology and Strength). The pathwise condition is sometimes called a sure or strong arbitrage, but it is not synonymous with *arbitrage of the first kind*. The latter is the condition dual to No Unbounded Profit with Bounded Risk (NUPBR/NA1) and is a distinct, weaker viability concept [21]. Classical no-arbitrage alone already excludes the universal strategy in Theorem 8; NFLVR is invoked below because it supplies the standard martingale-measure formulation.

B. NA, NUPBR, and Benchmark-Relative Gains

The relevant no-arbitrage notions do not form the simple chain sometimes suggested in informal discussions. In the standard semimartingale setting, NFLVR combines classical NA with NUPBR, whereas NA and NUPBR are in general logically distinct. NUPBR is equivalent, under standard portfolio hypotheses, to the existence of a numeraire portfolio whose positive relative-wealth processes are supermartingales [21, 22]. It may hold even when the stronger NFLVR condition and an ELMM fail.

This distinction prevents two overstatements. First, Theorem 8 uses only classical NA; NUPBR alone is not silently substituted for it. If a zero-cost gain were non-negative at all intermediate times and positive at terminal time, scaling would violate NUPBR, but terminal pathwise positivity together with only a strategy-specific lower bound does not by itself supply that scaling

argument. Second, benchmark and stochastic-portfolio approaches can generate gains *relative* to a numeraire or market portfolio without producing zero-cost positive gain on every path. Relative arbitrage in diverse equity markets and real-world benchmark pricing therefore do not contradict the theorem [23, 24].

C. The Martingale Constraint

For a locally bounded semimartingale price process, the Delbaen–Schachermayer form of the First Fundamental Theorem states that No Free Lunch with Vanishing Risk (NFLVR) is equivalent to the existence of an equivalent local martingale measure (ELMM) $\mathbb{Q} \approx \mathbb{P}$ under which $\bar{S}$ is a local martingale [25, 26]. For general unbounded semimartingales the corresponding formulation uses an equivalent σ -martingale measure [27]; the corollary below retains the locally bounded setting.

Corollary 9 (No Universal Zero-Cost Strategy). *If an ELMM $\mathbb{Q}$ exists, no admissible self-financing strategy with zero initial capital can have $G_T > 0$ on every path.*

Proof. Under $\mathbb{Q}$, the admissible stochastic integral $G = H \cdot \bar{S}$ is a local martingale bounded from below and therefore a supermartingale. Hence $\mathbb{E}_{\mathbb{Q}}[G_T] \leq G_0 = 0$. If $G_T > 0$ on every path, then equivalence gives $G_T > 0$, $\mathbb{Q}$ -a.s., and consequently $\mathbb{E}_{\mathbb{Q}}[G_T] > 0$, a contradiction. $\square$

The ELMM conclusion concerns discounted gains under the pricing measure $\mathbb{Q}$. It does not say that every risky strategy has non-positive expected return under the physical measure $\mathbb{P}$; positive physical expected returns may compensate risk.

D. Win Rate Is Not Expected Value

The probability of closing a trade at a profit can greatly exceed 1/2 even in a fair market. Optional stopping gives the precise distinction.

Proposition 10 (Asymmetric Barriers in a Fair Random Walk). *Let X_n be a simple symmetric random walk with $X_0 = 0$, and for positive integers a, b let*

$$\tau = \inf\{n \geq 0 : X_n = a \text{ or } X_n = -b\}.$$

Then

$$\mathbb{P}(X_\tau = a) = \frac{b}{a+b}, \quad \mathbb{E}[X_\tau] = 0. \quad (12)$$

Proof. The stopped process $X_{n \wedge \tau}$ is a bounded martingale, so Doob’s optional-stopping theorem applies [28]. If $p = \mathbb{P}(X_\tau = a)$, then $0 = \mathbb{E}[X_\tau] = ap - b(1 - p)$, which gives $p = b/(a + b)$. $\square$

With a take-profit of $a = 1$ and a stop-loss of $b = 9$, the win probability is 0.9, but the expected payoff is zero: nine frequent unit gains are offset by one nine-unit loss in expectation. Thus prediction accuracy, trade win rate, and expected discounted profit are three different quantities.

V. THE COMBINATORIAL PERSPECTIVE

Here the opposing side is an ensemble of market records rather than an adaptive market. The optimization NFL theorem and the elementary prediction result needed for market-direction claims should be distinguished.

Theorem 11 (Optimization NFL, Wolpert–Macready). *For finite search and cost spaces, when performance is averaged uniformly over all objective functions, any two non-revisiting black-box search algorithms have identical averaged performance [29].*

The theorem concerns oracle-based optimization over function spaces. It does not, by itself, imply a statement about self-financing gains on stochastic price paths. A sharpening makes its structural assumption explicit: for a uniform distribution on a finite function class, the NFL equality for all non-revisiting algorithms holds precisely when the class is closed under permutations of the search domain [30]. Financial path classes are generally not permutation-closed: temporal order, continuity, financing, and no-arbitrage restrictions distinguish one permutation from another. Thus a financial edge, when present, resides in exactly such symmetry-breaking structure and the inductive bias used to model it.

For binary directional prediction the relevant result is simpler and can be derived directly.

Proposition 12 (Uniform Binary Prediction). *Let $X_1, \dots, X_T$ be uniformly distributed on $\{0, 1\}^T$. A causal deterministic predictor chooses $\hat{X}_t = f_t(X_1, \dots, X_{t-1})$. Its mean accuracy*

$$A_T = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\hat{X}_t = X_t\}$$

satisfies $\mathbb{E}[A_T] = 1/2$. The same holds for a randomized predictor whose private randomization is independent of the next bit.

Proof. Conditional on every history $X_1, \dots, X_{t-1}$, the next bit X_t is an independent fair bit. Therefore $\mathbb{P}(\hat{X}_t = X_t \mid X_1, \dots, X_{t-1}) = 1/2$. Linearity of expectation gives the result. Conditioning additionally on the predictor’s private randomization proves the randomized case. $\square$

Consequently, no predictor can achieve expected directional accuracy greater than 1/2 under the uniform distribution over all binary sequences; in particular, none can exceed 1/2 on every sequence. This is an NFL symmetry statement, not a model of actual markets. Financial

sign sequences are not uniform in general, and a directional forecast does not determine trading profit because payoff sizes, financing, and stopping rules matter (Proposition 10).

The defensible practical conclusion is conditional: every repeatably successful predictor or strategy exploits non-uniform structure in the physical measure $\mathbb{P}$. This is the trader's indispensable insight or inductive bias—a trend, mean reversion, a risk premium, information, or an institutional constraint that breaks the NFL symmetry. Its edge is evidence about a restricted environment, not universal algorithmic superiority.

VI. ANALOGY TO SOCIAL CHOICE: SCOPE AND LIMITS

Arrow's Impossibility Theorem concerns social welfare functions on at least three alternatives. Under unrestricted preference profiles, collective rationality, Pareto unanimity, and independence of irrelevant alternatives, the aggregation rule must be dictatorial [31]. Its assumptions and objects differ from those of a trading model.

There is nevertheless a useful structural comparison:

Universal domain.: Arrow quantifies over all admissible preference profiles; pathwise trading universal-ly quantifies over all admissible price paths.

Jointly incompatible demands.: Arrow's fairness and consistency requirements cannot all be met by a non-dictatorial rule. In trading, zero-cost strict profit on every path is incompatible with no-arbitrage.

No theorem transfer.: Neither result proves the other. The parallel is a shared warning that unrestricted domains make apparently natural universal requirements inconsistent.

Yanofsky shows how many self-referential arguments can be organized using fixed points and fixed-point-free transformations [32]. The response in Eq. (3) anti-aligns returns with every nonzero position, but it admits the neutral fixed point $w_t = 0$. That fixed point yields zero gain rather than strict profit. The relevant obstruction is therefore the absence of a *strictly profitable* fixed response, not the absence of every fixed point.

VII. CASE STUDY: THE WHEEL OPTIONS STRATEGY

The Wheel sells a cash-secured put and, after assignment, sells a covered call against the acquired stock. Its economic exposure follows directly from put-call parity. Ignoring dividends and financing for the terminal-payoff identity,

$$S_T - (S_T - K)^+ = K - (K - S_T)^+. \quad (13)$$

The left-hand side is a covered-call payoff; the right-hand side is cash paying K plus a short put. After the corresponding time-zero costs are matched, a covered call and a cash-secured put are equivalent European payoffs. Rolling the Wheel therefore repeatedly maintains a put-like, short-downside-convexity exposure rather than creating directionless income. Equivalently,

$$\Phi_K(S_T) = S_T - (S_T - K)^+ = \begin{cases} S_T, & S_T < K, \\ K, & S_T \geq K, \end{cases} \quad (14)$$

which makes the one-for-one downside and capped upside explicit.

Option-selling returns may nevertheless have a positive physical mean. The difference between risk-neutral and physical expectations of volatility is a variance risk premium; option-market evidence documents compensation for bearing volatility and jump risk [33, 34]. This is an empirical, distribution-dependent explanation for some Wheel income, not a pathwise identity: the sign and magnitude depend on prices, strikes, transaction costs, and the physical law.

A. Failure I: Catastrophic Drop

If the underlying crashes after put assignment, the short call liability decreases, which benefits the call writer, but the loss on the stock dominates and the combined covered-call position can suffer a large net loss. Equation (13) makes this explicit: below K , the payoff falls one-for-one with S_T . A reversed rally/crash pair can be used as the stress test in Criterion 1, but time reversal alone is not the proof; the payoff identity is.

B. Failure II: Persistent Decline

Repeated option premiums offset part of a persistent decline but do not remove the renewed short-put exposure. The loss is path-dependent rather than generally linear: strikes, maturities, volatility, assignment dates, and position sizing determine the sequence of gains.

C. Failure III: Violent Breakout and Benchmark Underperformance

In a large rally, shares may be called away at the call strike. The Wheel can still realize a positive absolute profit, so opportunity cost alone is not a counterexample to strict positive P&L. It is instead a failure relative to a buy-and-hold benchmark: Eq. (13) caps the terminal payoff at K . No-arbitrage constrains the price of this payoff under $\mathbb{Q}$; it does not say that its physical expected return must be zero.

D. Failure IV: Absorbing Ruin Under Capital Constraints

A fully cash-secured, unlevered cycle has bounded loss, but an underlying that falls to zero can consume almost all committed capital net of premium. Leverage makes the absorbing nature of ruin explicit. If wealth evolves as $W_{t+1} = W_t(1 + fR_{t+1})$ and an admissible return satisfies $1 + fR_{t+1} = 0$, then $W_{t+1} = 0$; a self-financing strategy cannot recover without an external capital injection. This is a wealth-dynamics fact, not a consequence of undecidability.

Table I is a payoff-based classification, not a claim that each trajectory has positive probability under every continuous price model. Corollary 9 excludes strict positive discounted gain on all paths, but probabilities of particular failure classes require a specified physical measure.

Repeated option income also raises a time-versus-ensemble distinction. For multiplicative wealth $W_T = W_0 \prod_{t=1}^T (1 + fR_t)$, the realized per-period growth is

$$\frac{1}{T} \log \frac{W_T}{W_0} = \frac{1}{T} \sum_{t=1}^T \log(1 + fR_t),$$

not the logarithm of an ensemble-average payoff. Frequent small gains can coexist with poor long-run growth when rare losses are large, a point emphasized in discussions of non-ergodic economic growth [35]. This does not replace expected-utility analysis; it identifies which long-run observable is being optimized.

VIII. UNIVERSAL PORTFOLIOS AND DISTANCE FROM UNIVERSALITY

The word “universal” also appears in a celebrated positive theorem. For price-relative vectors $x_t \in \mathbb{R}_{>0}^m$, a constant-rebalanced portfolio b has wealth

$$S_T(b) = \prod_{t=1}^T b^\top x_t.$$

Cover’s universal portfolio is nonanticipating and, under the conditions of his theorem, has the same asymptotic exponential growth rate as the best constant-rebalanced portfolio chosen in hindsight on every individual bounded sequence [36]. In regret notation,

$$\frac{1}{T} \left[\log \max_b S_T(b) - \log S_T^{\text{UP}} \right] \rightarrow 0. \quad (15)$$

Equation (15) is a relative guarantee, not an absolute-profit guarantee. If every constant-rebalanced portfolio loses money on a path, the universal portfolio may lose as well while retaining vanishing per-period regret. Cover’s result therefore does not contradict Corollary 9; it illustrates the strongest useful meaning of “universal” once the benchmark class is specified.

a. A Hierarchy of Attainable Claims The useful alternatives to pathwise universality are not weaker points on one single order; they change the benchmark, horizon, or probability quantifier. Table II records the distinctions.

Superhedging duality gives another quantitative notion: once a contingent claim and an admissible model class are fixed, its superhedging price is the least initial capital needed for a pathwise guarantee. Requiring zero initial capital while demanding a strictly positive payoff is precisely the degenerate, arbitrage-producing endpoint ruled out above.

IX. FAPP STRATEGIES AND THEIR HONEST SCOPE

While pathwise universal profit is precluded, strategies may be profitable *for all practical purposes* (FAPP, following Bell [37]) relative to a physical measure $\mathbb{P}$, a benchmark, and a finite horizon. Market making may earn spreads when adverse selection and inventory risk are controlled, and Kelly investing maximizes expected logarithmic growth when the relevant distribution is sufficiently well specified [38]. Neither is an unconditional profit guarantee.

This is where “insight” acquires an operational meaning. A trader specifies a restricted market class and a mechanism expected to persist in it, then accepts that the claim is falsifiable outside that class. A lucky finite gain needs no insight; a repeatable edge, positive conditional expectation, or sustained benchmark outperformance does.

Economic equilibrium also leaves room for conditional edges. Grossman and Stiglitz show that perfectly informationally efficient markets are inconsistent with costly information acquisition: if prices revealed all information without reward, agents would not pay to produce that information [39]. Conversely, the Milgrom–Stokey no-trade theorem shows that, under common priors, common knowledge of rationality, risk aversion, and an initially Pareto-optimal allocation, private information alone cannot induce mutually agreeable non-null trade [40]. These are equilibrium boundaries, not substitutes for the FTAP.

a. When the Trader Changes the Market The separation between trader and market becomes endogenous when a strategy is large, crowded, or public. Its orders consume liquidity and move quotes; its signals invite imitation or anticipatory trading; and its success changes the distribution on which it was estimated. In that case the deployed price path is better written P^σ : it is partly generated by the use of σ itself. The analogy is an invasive classical measurement or feedback experiment, in which the probe perturbs the observed process. In finance the mechanisms are price impact and strategic adaptation, not a physical measurement postulate.

Lo’s Adaptive Markets Hypothesis [41] supplies a less

TABLE I. Wheel failure modes and the mathematical mechanism illustrating each one.

Failure	Trajectory Class	Relevant Result	Mechanism
I: Crash	Large downside move	Eq. (13)	Put-like downside
II: Bleed	Persistent negative drift	Repeated payoff exposure	Premiums do not insure the stock
III: Breakout	Large upside move	Eq. (13)	Capped benchmark return
IV: Ruin	Capital exhaustion	Wealth recursion	Zero wealth is absorbing

TABLE II. Different meanings of a successful or universal strategy.

Notion	Guarantee	Status
Pathwise universal profit	Positive discounted zero-cost gain on every admissible path	Excluded by classical NA
Relative arbitrage	Outperformance of a specified market or numeraire portfolio	Possible under structural market assumptions
Cover universality	Vanishing per-period log regret to the best constant-rebalanced portfolio	Pathwise but benchmark-relative
FAPP or statistical edge	Positive performance under a specified physical law, data protocol, and horizon	Conditional and empirically falsifiable

idealized interpretation. As capital enters a successful strategy, competition, price impact, and strategic response may erode its edge. The market need not become the perfect adversary of Theorem 1; adaptation merely makes the physical distribution endogenous to widespread strategy use. Nor does impact prove that every large strategy loses; it invalidates naive extrapolation from a backtest that assumes unlimited scale without changing the environment. Market selection also needs hypotheses: Sandroni shows conditions under which agents with inaccurate beliefs lose wealth share, while emphasizing that the result depends on savings and preference structure [42]. Selection is therefore an equilibrium mechanism, not an unconditional theorem that the most accurate forecaster always dominates.

Empirical strategy discovery adds a separate multiple-testing problem. Trying many specifications and reporting the best backtest makes its apparent Sharpe ratio an order statistic rather than an unbiased performance estimate. The Deflated Sharpe Ratio corrects for selection, non-normal returns, and the number of trials [43]. Such corrections do not prove an edge; they make a finite-sample FAPP claim more honestly falsifiable.

Automated execution can amplify feedback when many systems share triggers, funding constraints, or risk models. The 1987 portfolio-insurance cascade [44] and the 2010 Flash Crash [45] illustrate such interactions. Their relevance comes from market impact and coupled control rules, not from Rice’s theorem.

X. CONCLUSION

Everything returns to the same pair: a trader and a market. Fix a total deterministic trading program and a

responsive market can compute the trader’s announced position and reverse its intended effect. This is the diagonal counterpath and the precise quantifier reversal $\forall \sigma \exists \mathcal{E}_\sigma$: every such trader has a market on which it fails to win. If the trader stays flat, the result is zero; if it takes a nonzero position, the constructed market makes that trade lose.

Universal computation adds an independent rule-inference boundary. An arbitrary computation can be encoded either in the behavior of a trader on a fixed rising path or in a future event of a total computable market generator. A master program deciding eventual profit or every encoded market event would therefore decide the Halting Problem. Gold-style inductive inference places this explicit reduction in a broader learning setting. The claim is not that restricted market models cannot be learned, but that no computable learner or certifier covers the unrestricted computationally universal class.

At the passive extreme, a Martin-Löf-random price-direction record relative to a computable fair measure need not react to the trader at all. It passes every effective null test and blocks unbounded success by every lower-semicomputable nonnegative fair-game capital process. This is not the adaptive diagonal construction, and neither statement is deterministic chaos. Between those extremes, a sufficiently large, crowded, or public trader changes prices and other traders’ responses; the trader becomes part of the market it is trying to exploit.

The existing financial and combinatorial results sharpen, rather than bury, this core idea. Strict zero-cost discounted profit on every path is classical arbitrage. Uniform binary averaging fixes expected directional accuracy at 1/2, while optional stopping shows why a high trade win rate need not imply positive expected gain. The Wheel’s put-like payoff, Cover’s benchmark-relative guar-

antee, equilibrium information results, and FAPP strategies show what remains possible: conditional growth, relative outperformance, and transient informational or risk-bearing rents.

A trader can therefore win repeatably only conditionally, by possessing some model, information, or structural insight that narrows the market class. Time reversal is the simplest computable falsification probe of that insight; many other stress transformations are possible. Passing them supports a limited robustness claim, never an insight-free strategy that wins against every market. Any practical claim must state its benchmark, financing, admissibility constraints, horizon, data-selection protocol, and assumptions about how deployment changes the environment.

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OpenAI Codex (GPT-5) was used to assist with literature organization, LaTeX editing, and checks of derivations. The author specified the scientific direction and prompts; model output retained in the manuscript was checked against the cited sources and explicit calculations. The author accepts full responsibility for the final text.

DATA AVAILABILITY

This is a purely mathematical work, and no data or software were created or analyzed in this study.

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